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LONG WALL 09:

⇒ Excavation:

length = 13' + (4.5" + (3 x 2.25)") + 9"
= 14'8.25" = 14.6875"

width = 3'

height = 1'9" = 1.75'

volume = 77.11 bcf

⇒ Concrete:

length = 14.6875"

width = 3'

height = 9" = 0.75'

volume = 33.05 cft

⇒ Brickwork:

→ Step ①:

length = 14'8.25" - 2.25" = 14'6" = 14.5'

width = 2'3" = 2.25'

height = 6" = 0.5'

Volume = 16.3125 cft

Sunny®

→ Step ②:

length = 14'6" - 2·25" = 14'3·75" = 14.3125'

width = 1'10.5" = 1.875'

height = 6" = 0.5'

volume = 13.42 cft

→ Step ③:

length = 14'3.75" - 2·25" = 14'1·5" = 14.125'

width = 1'6" = 1.5'

height = 6" = 0.5'

volume = 10.594 cft

→ Step ④:

length = 14.125'

width = 1.125'

height = 0.5'

volume = 7.95 cft

→ Brickwork upto F.F.L

length = 14.125'

width = 9" = 0.75'

height = 2'6" = 2.5'

volume = 26.48 cft

Total brickwork = 74.7565 cft

Sunny

Back fill till N.G.L :

Volume = Excavation - (Brickwork till NGL + PCC)

= 77.11 - (62.7825)

= 14.3275 cft

SHORT WALL 13:

Excavation:

length = $(15'6'' - 9'') - ((2.25'' \times 4) + 4.5'')$

= $13'7.5'' = 13.625'$

width = $3'$

height = $9'' + (6'' \times 2) = 1'9'' = 1.75'$

vol = 71.5155 bcf

concrete:

length = $13.625'$

width = $3'$

height = $9'' = 0.75'$

volume = $30.65'$ cft

Brickwork:

step ①:

length = $13'7.5'' + 4.5'' = 14'$

width = $2'3'' = 2.25'$

height = $6'' = 0.5'$

volume = 15.75 cft

Step ②

length = 14' + 4.5" = 14'4.5" = 14.375'
width = 1'10.5" = 0.875 + 1 = 1.875'
height = 6" = 0.5'

volume = 13.4766 cft

Step ③:

length = 14'4.5" + 4.5" = 14'9" = 14.75'
width = 1'6" = 1.5'
height = 6" = 0.5'

volume = 11.0625 cft

Step ④:

length = 14'9" + 4.5" = 15'1.5" = 15.125'
width = 1'1.5" = 1.125'
height = 6" = 0.5'

volume = 8.51 cft

Brickwork till F.F.L:

length = 15'1.5" + 4.5" = 15'6" = 15.5'
width = 9" = 0.75'
height = 2'6" = 2.5'

volume = 29.0625 cft

SHORT WALL 14:

=> Excavation:

$$\text{length} = 6' - 4.5'' - 3(4.5)'' - 4.5'' - 4(2.25)''$$
$$= 40.5''$$

$$\text{width} = 9'' + (4.5 \times 2)'' = 18''$$

$$\text{height} = 9'' + 6'' + 6'' = 21''$$

$$\text{volume} = 15309 \text{ in}^3 = 8.8594 \text{ bcf}$$

=> Concrete:

length = 40.5" width = 18" height = 6"

Volume = 4374 in^3 = 2.53125 cft

→ Brickwork: Step ①:

length = 40.5" + 4.5" = 45" width = 13.5"

height = 3"

volume = 1.055 ccf

→ Step ②:

length = 45" + 4.5" = 49.5" width = 13.5"

height = 3"

volume = 1.1602 ccf

→ Step ③:

length = 49.5" + 4.5" = 54"

width = 9" height = 3"

volume = 54 × 9 × 3 = 1458 in^3 = 0.84375 ccf

→ Step ④:

length = 54" + ~~4.5~~ 2.25" = 56.25"

width = 9" height = 6"

volume = 1.758 ccf

→ Step ⑤:

length = 56.25" + 2.25 = 58.5"
width = 9" height = 6"
volume = 1.898 cft

→ Step ⑥:

length = 60.75"
width = 9" height = 6"

volume = 1.898 cft

→ Step ⑦:

height = 30" width = 9"
length = 63"

volume = 9.844 cft

Total Brickwork:

18.387 cft

Backfill till NSL:

Excavation - (Brickwork till NGL + concrete)
= 1.51172 ccf

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LONG WALL II:

⇒ Excavation:

length = 156"

width = 9" + 2(4.5)"

= 1'6"

height = 9" + 6" + 6" = 21" = 1'9"

volume = 58968 in^3

= 34.125 bcf

⇒ Concrete:

length = 156"

width = 18"

height = 6"

volume = 156 × 18 × 6 = 16848 in^3

= 9.75 cft

⇒ Brickwork:

→ Step ①:

length = 156"

width = 1'6" - 4.5" = 13.5"

height = 6"

volume = 7.3125 cft

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→ Step ②:

length = 156"
width = 9"
height = 51"

volume = 46.21875 cft

→ Step ③: (for NGL)

height = 9" length = 156" width = 9"

volume = 7.3125 cft

Total brickwork = 60.84375 cft

Backfill till NGL:

= Excavation − (Brickwork till NSL + PCC)
= 9.75 ccf

volume = 0.5625 cft

→ step (4):

length = 9" width = 18 - 4.5 = 13.5"

height = 6"

volume = 0.42187 cft

→ step (5):

length = 9" width = 9" height = 6"

volume = 1.40625 cft

Total Brickwork = 3.9375 cft

Backfill (fill NSL):

= 0.703125 cct

COLUMN C1 & C2:

⇒ Excavation:

length = 9" width = 36"
height = 9" + 6" + 6" = 21"

volume = 3.9375 cft

⇒ Concrete:

length = 9" width = 36" height = 9"
volume = 1.6875 cft

⇒ Brickwork:

→ step ①:

height = 6" width = 36" - 9" = 27"
length = 9"

volume = 0.703125 cft

→ step ②:

length = 9" width = 27 - 4.5 = 22.5"
height = 6"

volume = 0.703125 cft

→ step ③:

length = 9" width = 22.5 - 4.5 = 18"
height = 6"

LONG WALL 10: (Ramp AREA)

⇒ Excavation ⇒

length = 8'
width = 1'6" = 1.5'
height = 2.5'

volume = 22.30 bcf

⇒ Concrete:

length = 8'
width = 1.5'
height = 0.5

volume = 6 cft

⇒ Brickwork: step ①:

length = 8'
width = 1.125'
height = 0.5'

volume = 4.5 cft

⟶ step ②

length (upto top of RAMP)
length = 8' width = 9" = 0.75'

height = 2'
volume = 13.5 cft

total brick work = 13.5 + 4.5
= 18 c.ft

Back fill:

= excavation - (Brickwork + concrete)
= 30 - 20.625 = 9.375 cft